Micro I, class 14

We began a discussion of choices under uncertainty. We will consider two approaches to this topic: one, associated with von Neumann and Morgenstern (1944), assumes that the choice set is the set of *probability distributions over outcomes*; the other, associated with Leonard Savage (1954), assumes that the choice set is the set of *acts* (mappings from a set of unknown states of the world into outcomes). Of the second, more later.

The details are in my notes on Choice under Uncertainty, but the primitives of the approach associated with von Neumann-Morgenstern are a nonempty finite set X of *outcomes*; a set $\mathcal{L}$ of *probability distributions* on X; and a *binary relation* $\succsim \subset \mathcal{L} \times \mathcal{L}$. Possible axioms on $\succsim$ are *completeness*; *transitivity*; *mixture continuity*; and *independence*. The first three you've seen before, though independence *seems* to be new.

Indeed, MWG write

> The independence axiom is at the heart of the theory of choice under uncertainty. It is unlike *anything* [emphasis added] encountered in the formal theory of preference-based choice discussed in Chapter 1 or its applications in Chapters 3-5. This is so precisely because it exploits, in a fundamental manner, the structure of uncertainty present in the model. In the theory of consumer demand, for example, there is no reason to believe that a consumer's preferences over various bundles of goods 1 and 2 should be independent of the quantities of the other goods that he will consume. In the present context, however, it *is* natural to think... (p. 172).

And they go on to write, in our notation, that it is natural to think that the preference between q and p, elements of $\mathcal{L}$, should determine preference between $\alpha q + (1-\alpha)r$ and $\alpha p + (1-\alpha)r$, when the convex combinations are interpreted as *two*-stage distributions; this is so, they argue, since, if the first stage reveals that the first element in the convex combination will determine the outcome (p or q), the common second element, r, is irrelevant (and if the first-stage reveals that the second element, r in both cases, determines the outcome, then the p or q is irrelevant.)[1]

I have two comments on this passage. First, since you have worked MWG 3.G.4, you know that you have encountered something *very* like the independence axiom in their Chapter 3. The axiom you encountered there asserts precisely that conditional preference over any subset of goods is independent of the quantities of the other goods.[2] The second is this: commodities can be distinguished, not just by date and location, but also by the uncertain state of the world under which they will be available (as MWG acknowledge on page 12). This observation is an important one for the Arrow-Debreu-McKenzie model of general equilibrium.[3] Specifically, "the formal theory of preference-based choice discussed in Chapter 1 or its applications in Chapters 3-5" includes *as a special case* a theory of choice under uncertainty, to which the axiom in MWG 3.G.4 might or might not be applied.

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[1] This argument seems to have originated in Samuelson, Paul A. "Probability, utility, and the independence axiom." *Econometrica* (1952): 670-678.

[2] The best exposition of the relationship between the axiom in 3.G.4—sometimes called *coordinate independence*—and the independence axiom is Fishburn, Peter, and Peter Wakker. "The invention of the independence condition for preferences." *Management Science* 41.7 (1995): 1130-1144.

[3] See Arrow, Kenneth J. "Le role des valeurs boursieres pour la repartition la meilleure des risques," *Econometrie*, (1953) 41-47, English translation as "The role of securities in the optimal allocation of risk-bearing." *Review of Economic Studies* 31 (1964): 91-96; and Chapter 7 in Debreu, Gerard. *Theory of value: An axiomatic analysis of economic equilibrium.* Yale University Press, 1959. Debreu's Chapter 7 is a reworking of his "Uné Économie de l'Incertain." Working Paper, Electricité de France, 1953.